

Q #1

Multiple Select Type

Award: 1

Penalty: 0

Machine Learning

We consider two regression models:

- Model A: $\hat{y} = \theta_0 + \theta_1 x$
- Model B: $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

Here:

- x_1 in Model B is the same feature as x in Model A,
- x_2 in Model B is an additional new feature not present in Model A.

We fit both models using the same dataset, and we fit both models using all data available to us, with coefficients chosen by minimising average squared loss (MSE).

Let

- MSE_A, MSE_B : mean squared errors of Model A and Model B, respectively,
- $\hat{\theta}_{1,A}, \hat{\theta}_{1,B}$: the estimated coefficients of x (or x_1) in the two models.

Which of the following statements are guaranteed to be true?

A. $MSE_A \geq MSE_B$

B. $MSE_A \leq MSE_B$

C. $\hat{\theta}_{1,A} = \hat{\theta}_{1,B}$

D. $\hat{\theta}_{1,A} \neq \hat{\theta}_{1,B}$

Q #2

Multiple Select Type

Award: 1

Penalty: 0

Machine Learning

Suppose we have $n = 4$ observations, and our model includes an intercept term. The residual vector is defined as

$$e = y - X\hat{\theta}$$

Which of the following are possible valid residual vectors? (Select all that apply.)

A. $e = [-1, 2, 3, -4]^T$

B. $e = [-1, 2, 3, 4]^T$

C. $e = [1, -1, 1, -1]^T$

D. $e = [0, 0.1, 0, 0.1]^T$

Q #3

Multiple Select Type

Award: 1

Penalty: 0

Machine Learning

Suppose we fit a linear regression model with an intercept term. Let the residual vector be

$$e = y - X\hat{\theta}$$

Which of the following statements must always hold?

A. The residuals are orthogonal to the fitted values $\hat{y}$.

B. The sum of residuals is zero.

C. The residuals are orthogonal to each column of X .

D. At least one residual must be zero if an intercept is included.

Q #4

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Congratulations! You've just landed a new job as a data scientist at Corn Hole Inc., which runs a training program for cornhole (a competitive party game with bean bags). Novices practice for a given number of hours and then play a scrimmage game totalling 100 points. You decide to model a linear relationship between a novice's practice time x and their scrimmage score y . Your model is $\hat{y} = \theta x$, where θ is a scalar.

Rather than using L_1 or L_2 regularization, you propose a new regularization term to penalize models with large magnitudes of θ . Which choice is most appropriate?

A. $e^{-\theta^2}$

B. $5^{|\theta|}$

C. $\frac{1}{\theta}$

D. θ^3

Q #5

Numerical Type

Award: 1

Penalty: 0

Machine Learning

We use the constant model $\hat{y} = \theta$ and the weighted mean squared error (WMSE)

$$\text{WMSE} = \frac{1}{n} \sum_{i=1}^n w_i (y_i - \theta)^2.$$

For the data

i	x_i	y_i	w_i
1	100	10	0.5
2	50	20	1.0
3	30	30	1.5
4	20	10	1.0

what is WMSE when $\theta = 20$?

Q #6

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

For the constant model $\hat{y} = \theta$ and the weighted mean squared error (WMSE)

$$\text{WMSE} = \frac{1}{n} \sum_{i=1}^n w_i (y_i - \theta)^2,$$

what is $\frac{\partial \text{WMSE}}{\partial \theta}$?

- A. $\frac{2}{n} \sum_{i=1}^n w_i (y_i - \theta)$
- B. $-\frac{2}{n} \sum_{i=1}^n w_i (y_i - \theta)$
- C. $\frac{1}{n} \sum_{i=1}^n w_i (y_i - \theta)$
- D. $-\frac{1}{n} \sum_{i=1}^n w_i (y_i - \theta)$

Q #7

Multiple Select Type

Award: 1

Penalty: 0

Machine Learning

Rayna brings us some other models to consider. Which of the following are linear in the parameters?

Select all that apply.

- A. $\hat{y}_i = \theta_0 + \theta_1 x_i$
- B. $\hat{y}_i = \theta_0 e^{x_i}$
- C. $\hat{y}_i = \theta_0 \theta_1 x_i$
- D. $\hat{y}_i = \theta_0 + \ln(x_i) \theta_1$

Q #8

Multiple Select Type

Award: 1

Penalty: 0

Machine Learning

In general, which of the following statements are true? Select all that apply.

- A. Initial parameter values do not affect the gradient-descent convergence.
- B. The learning rate affects the speed of gradient descent and whether the model converges.
- C. Negative values are an appropriate choice for the learning rate.
- D. Each update of θ may jump back and forth across the two sides of the optimal θ .

Q #9

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Write down a closed-form solution for the optimal parameters w that minimize the loss function

$$L(w) = \sum_{i=1}^n (y_i - x_i^T w)^2 + \lambda \|w\|_2^2$$

in terms of (1) the $n \times d$ matrix X whose i -th row is a $1 \times n$ vector x_i^T (a sample), (2) the $n \times 1$ vector y whose i -th entry is y_i , and (3) the scalar λ .

(You may assume that any relevant matrix is invertible.)

- A. $\hat{w} = (X^T X)^{-1} X y + \lambda I$
- B. $\hat{w} = 2(X^T X + \lambda I)^{-1} X^T y$
- C. $\hat{w} = \lambda(X^T X)^{-1} X^T y$
- D. $\hat{w} = (X^T X + \lambda I)^{-1} X^T y$

Q #10

Multiple Choice Type

Award: 1

Penalty: 0.33

Machine Learning

Select the correct option about regression with L2 regularization (also called Ridge Regression).

- a. The Ridge regression technique prevents coefficients from rising too high.
 - b. As $\lambda \rightarrow \infty$, the impact of the penalty grows, and the ridge regression coefficient estimates will approach infinity.
- A. Both statements are true.
- B. Both statements are false.
- C. Statement *a* is true, Statement *b* is false.
- D. Statement *b* is true, Statement *a* is false.

Q #11

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

We fit a linear regression model $\hat{y} = X\theta$ with residuals $e = y - \hat{y}$. If the first column of X is all ones (bias term), the following holds:

1. $X^T e = 0$
2. $\mathbf{1}^T e = 0$, where $\mathbf{1}$ is a vector of ones
3. $e \perp \hat{y}$
4. $\sum_{i=1}^n e_i = 0$

If the bias term is removed, which statements no longer hold?

- A. Only (1)
- B. (2) and (4)
- C. (3) only
- D. (2), (3), and (4)

Q #12

Multiple Select Type

Award: 2

Penalty: 0

Machine Learning

Consider a multiple linear regression model

$$\hat{y} = X\theta$$

with squared error loss and the following exponential regularization term:

$$R(\theta) = \lambda \sum_{i=1}^p 10^{\theta_i}$$

The objective function is

$$\frac{1}{n} \|Y - X\theta\|_2^2 + R(\theta)$$

Which of the following statements are correct? (Select all that apply.)

- A. The optimal solution is the ordinary least squares solution $(X^T X)^{-1} X^T Y$.
- B. The optimal solution is $(X^{-1} + \lambda I) X^T Y$.
- C. Gradient descent can be used to compute the optimal solution.
- D. λ represents the regularization coefficient.

Q #13

Multiple Select Type

Award: 2

Penalty: 0

Machine Learning

Rahul is unhappy with the multiple linear regression model's performance, and he decides to experiment with using Ridge regression ($\lambda > 0$) to predict the number of movies per year. Which of the following statements is true if he uses stochastic gradient descent to compute $\hat{\theta}$?

- A. Using a fixed number of iterations may lead to non optimal solution
- B. There isn't a unique optimal solution to minimize the Ridge objective function.
- C. Given a good choice of λ , the Ridge regression model can reduce overfitting.
- D. Stochastic gradient descent reduces the objective function value at every iteration.

Q #14

Multiple Choice Type

Award: 2

Penalty: 0.67

Machine Learning

We fit a simple linear regression using the dataset

x	y
0	1
2	6
0	6
1	6
2	1

The parameters are initialized as $\theta^{(0)} = [0, 0]$ and the learning rate is $\alpha = 0.1$. The objective function is

$$\hat{y} = a + bx \quad R(\theta) = \frac{1}{2} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

After one gradient descent update, what is

$$\theta^{(1)} = [a^{(1)}, b^{(1)}]?$$

- A. $[0, 0]$
- B. $[2, 2]$
- C. $[-2, -2]$
- D. $[1, 1]$

Q #15

Numerical Type

Award: 2

Penalty: 0

Machine Learning

You want to use gradient descent (GD) to fit a model with the loss function

$$L(\theta_0) = (y - \theta_0)^4$$

Given $y = 1$, learning rate $\alpha = \frac{1}{4}$, and starting value $\theta_0^{(0)} = 0$, what will be the value of $\theta_0^{(1)}$ after the first GD update?

Your Answer:

Correct Answer: 1

Not Attempted

Time taken: 00min 00sec

Discuss